

Optimising Response-Adaptive Randomisation in Group Sequential Trials

A Bayesian decision-theoretic approach

Corey Voller

Supervisors: Haiyan Zheng & Christopher Jennison
Department of Mathematical Sciences, University of Bath

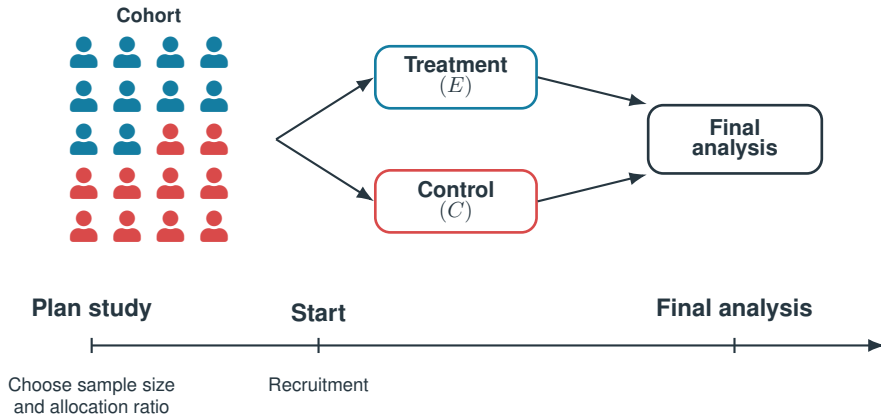
ISCB GMDS

Freiburg, Germany

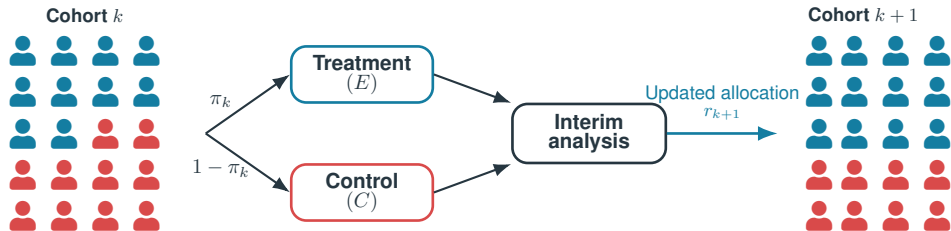
September 2026



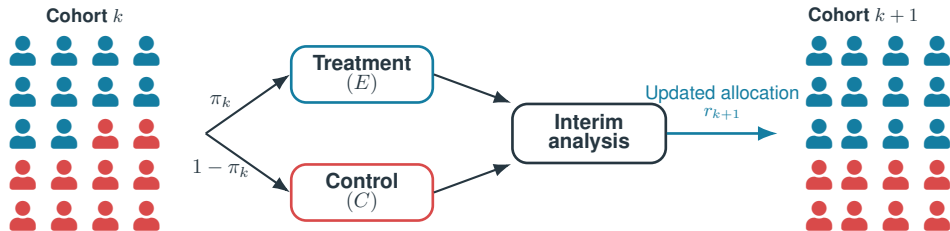
Basic trial design



Response-adaptive randomisation



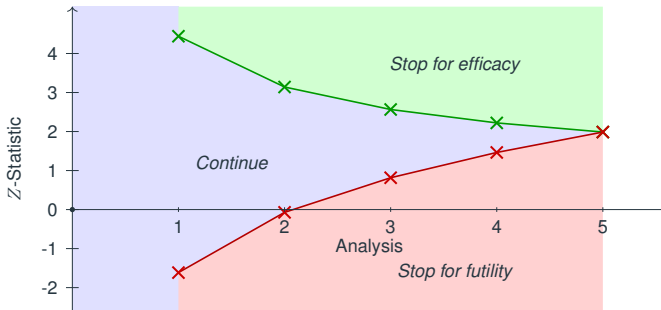
Response-adaptive randomisation



How should we choose the updated allocation?

Group sequential tests

A typical boundary for a one-sided test with test statistics $Z_1, \dots, Z_k$ is:

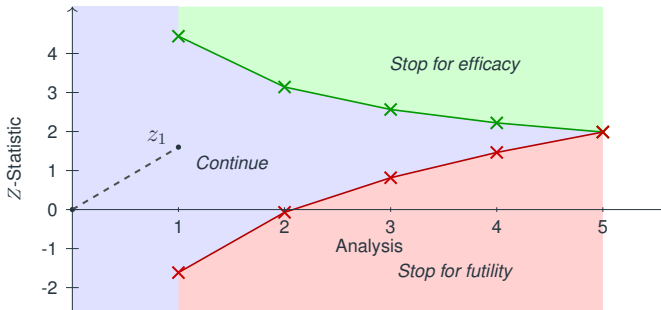


Crossing the upper boundary results in early stopping for a positive outcome, rejecting H_0 in favour of $\theta > 0$.

Crossing the lower boundary results in early stopping for futility, as there is insufficient evidence of benefit.

Group sequential tests

A typical boundary for a one-sided test with test statistics $Z_1, \dots, Z_k$ is:

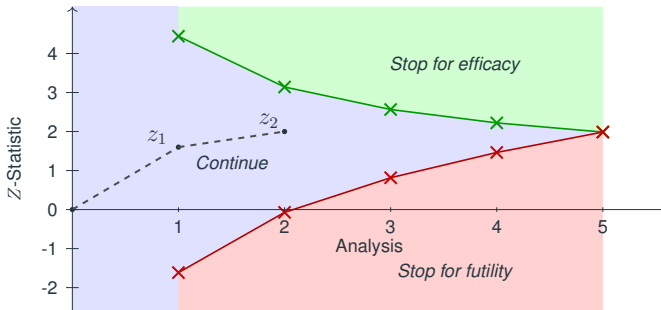


Crossing the upper boundary results in early stopping for a positive outcome, rejecting H_0 in favour of $\theta > 0$.

Crossing the lower boundary results in early stopping for futility, as there is insufficient evidence of benefit.

Group sequential tests

A typical boundary for a one-sided test with test statistics $Z_1, \dots, Z_k$ is:

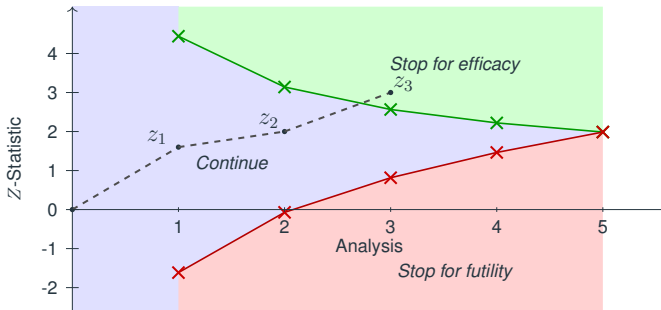


Crossing the upper boundary results in early stopping for a positive outcome, rejecting H_0 in favour of $\theta > 0$.

Crossing the lower boundary results in early stopping for futility, as there is insufficient evidence of benefit.

Group sequential tests

A typical boundary for a one-sided test with test statistics $Z_1, \dots, Z_k$ is:



Crossing the upper boundary results in early stopping for a positive outcome, rejecting H_0 in favour of $\theta > 0$.

Crossing the lower boundary results in early stopping for futility, as there is insufficient evidence of benefit.

Trial set up

Two-arm Phase III trial monitored at K planned analyses.

Treatment effect

$$\theta = \mu_E - \mu_C$$

Hypothesis

$$H_0 : \theta \leq 0 \quad \text{vs} \quad H_1 : \theta > 0$$

With

$$\alpha = 0.025, \quad 1 - \beta = 0.90 \quad \text{at} \quad \theta = \delta$$

where δ is a clinically relevant treatment difference

K planned analyses

accumulating information



$\hat{\theta}_k$: cumulative treatment-effect estimate

$$I_k = \left\{ \text{Var}(\hat{\theta}_k) \right\}^{-1}$$

information available at analysis k

Sequential distribution theory

The monitoring statistic is

$$Z_k = \hat{\theta}_k \sqrt{I_k},$$

Under the canonical joint distribution,

$$Z_k \sim N\left(\theta \sqrt{I_k}, 1\right), \quad k = 1, \dots, K,$$

and

$$\text{Cov}(Z_{k_1}, Z_{k_2}) = \sqrt{I_{k_1}/I_{k_2}}, \quad k_1 < k_2.$$

If the planned information levels $I_1, \dots, I_K$ are preserved, then $Z_1, \dots, Z_K$ retain the canonical joint distribution used to calibrate the stopping boundaries.

Group sequential designs and response-adaptive randomisation

Statistical Science
2023, Vol. 38, No. 2, 219–223
<https://doi.org/10.1214/23-STS865D>
Main article: <https://doi.org/10.1214/22-STS865>
© Institute of Mathematical Statistics, 2023

Comment: Group Sequential Designs with Response-Adaptive Randomisation

Christopher Jennison

Abstract. Group sequential Phase III trial designs enable early stopping for positive or negative study outcomes. Response-adaptive randomisation can be included in such designs with the sampling ratio in each group of subjects determined by the current treatment effect estimate. We demonstrate the potential of adaptive randomisation to reduce the number of patients receiving the inferior treatment, even when there is a delay in observing each patient's response. We also observe that using a fixed but unequal sampling ratio may offer a simpler way to achieve the same objectives.

Key words and phrases: Response-adaptive randomisation, group sequential test, inferior treatment number.

Loss-based allocation

Jennison (2023) and Jennison and Turnbull (2025) describe a group sequential design that aims to minimise the loss function:

$$L(\theta) = \begin{cases} I_E + a^{\theta/\delta} I_C & \text{if } \theta \geq 0 \\ I_C + a^{-\theta/\delta} I_E & \text{if } \theta \leq 0 \end{cases}$$

where a is a chosen constant. At each analysis we require

$$I_k = \{1/I_E^k + 1/I_C^k\}^{-1}$$

The ratio that minimises $L(\theta)$ subject to I_k is

$$r = \frac{I_E}{I_C} = a^{\theta/(2\delta)} \quad \text{equivalently,} \quad r = \frac{N_E}{N_C}.$$

Loss-based allocation

Jennison (2023) and Jennison and Turnbull (2025) describe a group sequential design that aims to minimise the loss function:

$$L(\theta) = \begin{cases} I_E + a^{\theta/\delta} I_C & \text{if } \theta \geq 0 \\ I_C + a^{-\theta/\delta} I_E & \text{if } \theta \leq 0 \end{cases}$$

where a is a chosen constant. At each analysis we require

$$I_k = \{1/I_E^k + 1/I_C^k\}^{-1}$$

The ratio that minimises $L(\theta)$ subject to I_k is $\hat{\theta}_{k-1}^{\text{MLE}}$

$$r = \frac{I_E}{I_C} = a^{\theta/(2\delta)} \quad \text{equivalently,} \quad r = \frac{N_E}{N_C}.$$

Loss-based allocation

Jennison (2023) and Jennison and Turnbull (2025) describe a group sequential design that aims to minimise the loss function:

$$L(\theta) = \begin{cases} I_E + a^{\theta/\delta} I_C & \text{if } \theta \geq 0 \\ I_C + a^{-\theta/\delta} I_E & \text{if } \theta \leq 0 \end{cases}$$

where a is a chosen constant. At each analysis we require

$$I_k = \{1/I_E^k + 1/I_C^k\}^{-1}$$

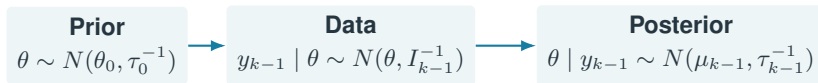
The ratio that minimises $L(\theta)$ subject to I_k is

$$\hat{\theta}_{k-1}^{\text{MLE}}$$

Early interim estimates
can be noisy

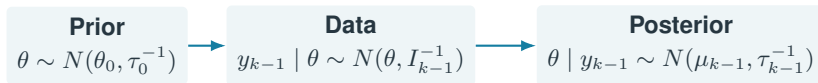
$$r = \frac{I_E}{I_C} = a^{\theta/(2\delta)} \quad \text{equivalently,} \quad r = \frac{N_E}{N_C}.$$

Bayesian allocation rules

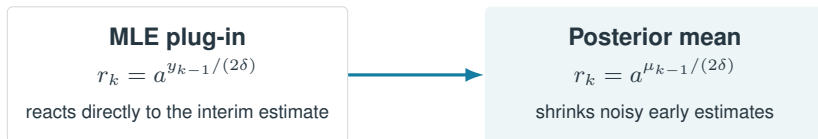


$$\mu_{k-1} = \frac{\tau_0 \theta_0 + I_{k-1} y_{k-1}}{\tau_0 + I_{k-1}}, \quad \tau_{k-1} = \tau_0 + I_{k-1}.$$

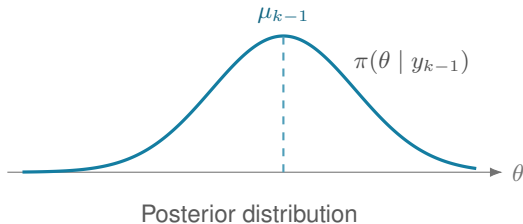
Bayesian allocation rules



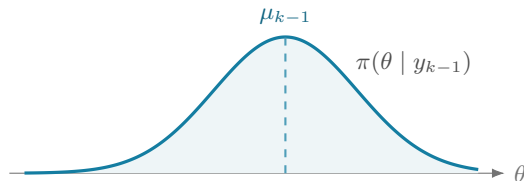
$$\mu_{k-1} = \frac{\tau_0 \theta_0 + I_{k-1} y_{k-1}}{\tau_0 + I_{k-1}}, \quad \tau_{k-1} = \tau_0 + I_{k-1}.$$



Bayesian allocation rules



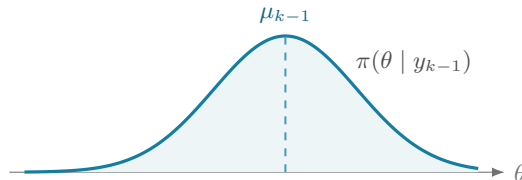
Bayesian allocation rules



Posterior integrated

$$r_k^{\text{PI}} = \arg \min_r \int L\{\theta; I_E(r), I_C(r)\} \pi(\theta \mid y_{k-1}) d\theta$$

Bayesian allocation rules



Posterior integrated

$$r_k^{\text{PI}} = \arg \min_r \int L\{\theta; I_E(r), I_C(r)\} \pi(\theta | y_{k-1}) d\theta$$

Stop-weighted posterior

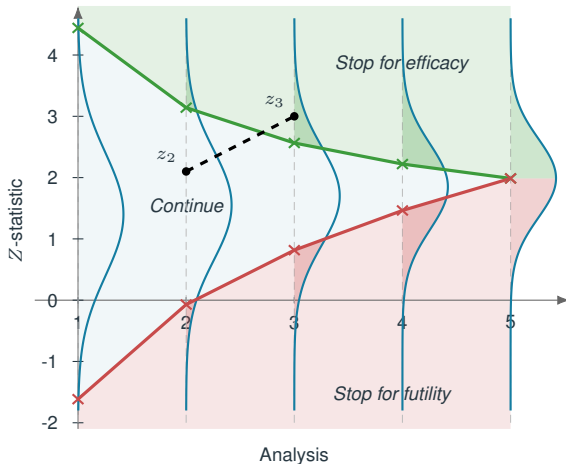
$$r_k^{\text{SW}} = \arg \min_r \int p_k(\theta; z_{k-1}) L\{\theta; I_E(r), I_C(r)\} \pi(\theta | y_{k-1}) d\theta$$

Probability of stopping at the next analysis

At analysis $k - 1$, conditional on the current result z_{k-1} and a candidate value of θ ,

The probability of stopping at the next analysis is

$$p_k(\theta; z_{k-1}) = P(Z_k < a_k \mid z_{k-1}, \theta) + P(Z_k > b_k \mid z_{k-1}, \theta)$$



Simulation study

Design

- ▶ Normal outcomes
- ▶ $K = 5$ analyses
- ▶ One-sided $\alpha = 0.025$
- ▶ 90% power at $\theta = \delta$
- ▶ Fixed planned information sequence

$$I_1, \dots, I_K$$

- ▶ 20 million replicates

Treatment-effect distribution

$$\theta \sim \pi_{\text{sim}}(\theta)$$

Prior distribution chosen such that

$$P(0 < \theta < \delta) = 0.95.$$

Primary performance measure

$$\mathbb{E}[L(\theta)]$$

Lower expected loss indicates better allocation of information between treatment and control.

Simulation Study

Method	No early stopping	Early stopping
Fixed, 1:1	313.595 (0.017)	240.989 (0.021)
RAR, MLE	304.978 (0.014)	234.938 (0.019)
Fixed, optimal ratio	303.106 (0.015)	232.609 (0.020)
RAR, posterior mean	302.291 (0.014)	232.313 (0.020)
RAR, posterior integrated	302.264 (0.014)	232.298 (0.020)
RAR, stop-weighted posterior	—	232.249 (0.020)
Oracle benchmark	300.785 (0.014)	231.009 (0.020)

Loss is reported as mean realised loss (MCSE). Results are based on 20 million replicates.

Conclusions

Bayesian methods **stabilise early allocation decisions**, reducing the impact of noisy interim estimates.

Using the whole posterior and accounting for stopping provides **additional, but increasingly modest, gains**.

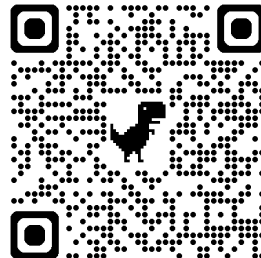
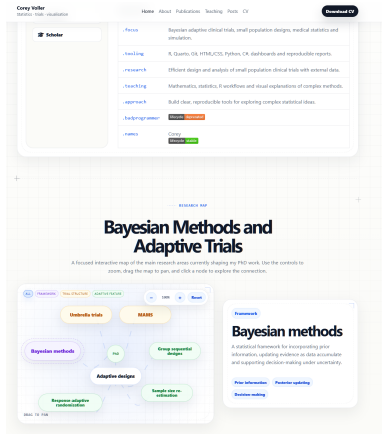
A well-chosen fixed unequal allocation can be **highly competitive**, while being considerably simpler to implement.

Group sequential stopping remains **the largest driver of the overall reduction in expected loss**.

References

-  Jennison, Christopher (May 1, 2023). “Comment: Group Sequential Designs with Response-Adaptive Randomisation”. In: *Statistical Science* 38.2. DOI: 10.1214/23-STS865D.
-  Jennison, Christopher and Bruce W. Turnbull (Oct. 2025). *Group Sequential and Adaptive Methods for Clinical Trials*. en. 2nd ed. Boca Raton: Chapman and Hall/CRC. DOI: 10.1201/9781584888482.

Slides



cgvoller.github.io

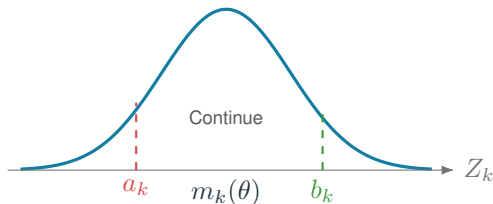
Appendix: Stopping probability at the next analysis

At analysis $k - 1$, conditional on the current result z_{k-1} and a candidate value of θ ,

$$m_k(\theta) = \frac{z_{k-1} \sqrt{I_{k-1}} + \theta(I_k - I_{k-1})}{v_k^2 = \frac{I_k - \sqrt{I_k}}{I_k}},$$

$$Z_k \mid (Z_{k-1} = z_{k-1}, \theta) \sim N(m_k(\theta), v_k^2).$$

$$p_k(\theta; z_{k-1}) = P(\text{stop at analysis } k \mid z_{k-1}, \theta).$$



Appendix: Stopping probability at the next analysis

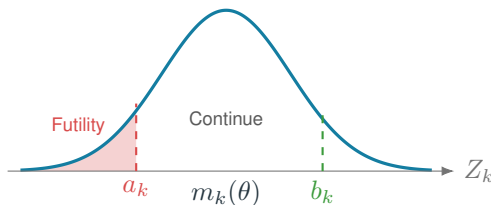
At analysis $k - 1$, conditional on the current result z_{k-1} and a candidate value of θ ,

$$m_k(\theta) = \frac{z_{k-1} \sqrt{I_{k-1}} + \theta(I_k - I_{k-1})}{v_k^2 = \frac{I_k - \sqrt{I_k}}{I_k}},$$

$$Z_k \mid (Z_{k-1} = z_{k-1}, \theta) \sim N(m_k(\theta), v_k^2).$$

$$p_k(\theta; z_{k-1}) = P(Z_k < a_k \mid z_{k-1}, \theta) + P(Z_k > b_k \mid z_{k-1}, \theta).$$

The stopping probability is the area in the two tails.



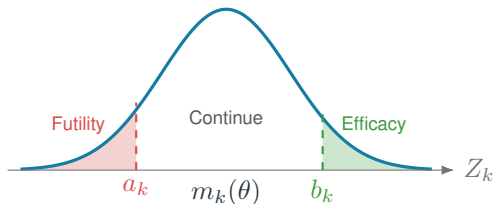
Appendix: Stopping probability at the next analysis

At analysis $k - 1$, conditional on the current result z_{k-1} and a candidate value of θ ,

$$m_k(\theta) = \frac{z_{k-1} \sqrt{I_{k-1}} + \theta(I_k - I_{k-1})}{v_k^2 = \frac{I_k - \sqrt{I_k}}{I_k}},$$

$$Z_k \mid (Z_{k-1} = z_{k-1}, \theta) \sim N(m_k(\theta), v_k^2).$$

$$p_k(\theta; z_{k-1}) = P(Z_k < a_k \mid z_{k-1}, \theta) + P(Z_k > b_k \mid z_{k-1}, \theta).$$



The stopping probability is the area in the two tails.